Chapter 13: Temperature, Kinetic Theory, and the Gas Laws

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201

Lecture 13

Part I

Temperature

Outline of Part I: Temperature

Temperature as Average Kinetic Energy

Temperature Scales

What is Temperature?

Temperature (T) — SI Unit: kelvin (K)

Temperature is proportional to the **average internal kinetic energy** of the atoms and molecules that compose a substance.

$$T \propto \overline{KE}_{\text{internal}}$$

- $\overline{KE}_{\text{internal}}$ is the *average* kinetic energy of the molecules that compose a substance.
- A hotter object has molecules moving faster on average.
- Temperature is a **scalar** quantity — it has no direction.

Temperature vs. Heat

Temperature

- A property of the object itself.
- Related to the **average** KE of molecules.
- Measured in °C, °F, or K.

Heat

- Energy transferred between objects.
- Flows from higher T to lower T.
- Measured in joules (J).

A spark from a fire is *hotter* than boiling water (higher T), but a pot of boiling water transfers *more heat* to your hand than a spark because it contains far more molecules.

Thermal Equilibrium & the Zeroth Law

Thermal Equilibrium

Two objects in contact eventually reach the **same temperature**; no net heat flows between them.

Zeroth Law of Thermodynamics

If system A is in thermal equilibrium with system B, and system B is in thermal equilibrium with system C, then system A is also in thermal equilibrium with system C.

- The Zeroth Law is the logical basis for **thermometers**: a thermometer (B) in equilibrium with a patient (A) reads the patient's temperature because B and A share the same T.

Outline of Part I: Temperature

Temperature Scales

Fahrenheit	Celsius	Kelvin
Water freezes: $32\,^\circ\mathrm{F}$	Water freezes: $0\,^\circ\mathrm{C}$	Water freezes: $273.15\,\mathrm{K}$
Water boils: $212\,^\circ\mathrm{F}$	Water boils: $100\,^\circ\mathrm{C}$	Water boils: $373.15\,\mathrm{K}$
Body temp: $98.6\,^\circ\mathrm{F}$	Body temp: $37.0\,^\circ\mathrm{C}$	Body temp: $310.15\,\mathrm{K}$
Absolute zero: $-459.67\,^\circ\mathrm{F}$	Absolute zero: $-273.15\,^\circ\mathrm{C}$	Absolute zero: $0\,\mathrm{K}$

Absolute Zero

Absolute Zero: $T = 0\,\mathrm{K}$

The temperature at which $\overline{KE}_{\text{internal}} = 0$.

If $KE = 0$, there is no molecular motion $\Rightarrow$ $0\,\mathrm{K}$ is the lowest possible temperature.

Absolute Zero is Not Attainable

- Atoms vibrate, rotate, and translate even at very low temperatures.
- Quantum mechanics forbids atoms from being completely at rest.
- $\therefore\ T = 0\,\mathrm{K}$ can be approached but never reached.

Why Kelvin?

- Kelvin is an **absolute** scale: $0\,\mathrm{K}$ means zero kinetic energy.
- $0\,°\mathrm{C}$ and $0\,°\mathrm{F}$ are arbitrary reference points — they do *not* imply zero kinetic energy.
- Always use Kelvin in physics equations.

Temperature Conversion Equations

Between Fahrenheit and Celsius

$$T_F = 1.8\,T_C + 32 \qquad \Longleftrightarrow \qquad T_C = \frac{T_F - 32}{1.8}$$

Between Celsius and Kelvin

$$T_K = T_C + 273.15 \qquad \Longleftrightarrow \qquad T_C = T_K - 273.15$$

- One Celsius degree = 1.8 Fahrenheit degrees (the scales differ in *size* of degree and *zero point*).
- One kelvin = one Celsius degree (the scales differ only in *zero point*).

Example: Body Temperature Conversion

Normal human body temperature is $37.0\,°\mathrm{C}$.
Express this in (a) Fahrenheit and (b) Kelvin.

(a)

$$T_F = 1.8\,T_C + 32$$

$$T_F = 1.8\,(37.0\,°\mathrm{C}) + 32$$

$$\boxed{T_F = 98.6\,°\mathrm{F}}$$

(b)

$$T_K = T_C + 273.15$$

$$T_K = 37.0\,°\mathrm{C} + 273.15$$

$$\boxed{T_K = 310.15\,\mathrm{K}}$$

Check Your Understanding

A patient arrives at the hospital with a fever of 104 °F.

(a) What is this temperature in Celsius?

(b) What is this temperature in Kelvin?

Come to lecture to see the answer.

Check Your Understanding

When a cold alcohol thermometer is placed in a hot liquid, the alcohol column *drops* slightly before rising.

Explain why this happens.

Come to lecture to see the answer.

Part II

Thermal Expansion

Outline of Part II: Thermal Expansion

Linear Expansion

Volume Expansion

Outline of Part II: Thermal Expansion

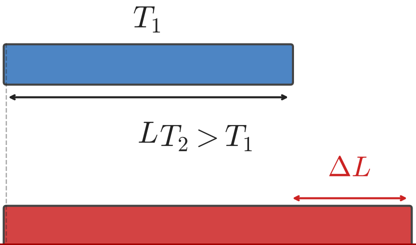
Linear Expansion

Volume Expansion

Why Do Objects Expand When Heated?

When temperature increases, atoms gain kinetic energy and vibrate more vigorously.

The average separation between atoms increases because the interatomic potential energy curve is *asymmetric* — atoms spend more time farther apart.



Real-World Examples

- Expansion joints in bridges and roads
- Slack in overhead power lines
- Bimetallic strip thermostats
- Loose-fitting jar lids (hot water expands the metal lid)

Linear Thermal Expansion

Linear Expansion — $\Delta L = \alpha L \Delta T$

$$\Delta L = \alpha L \Delta T$$

$$L_2 = L_1(\alpha \Delta T + 1)$$

- ΔL: change in length (m)
- L: original length (m)
- α: coefficient of linear expansion $\left(\frac{1}{\mathrm{K}} \text{ or } \frac{1}{{}^\circ\mathrm{C}}\right)$
- $\Delta T = T_2 - T_1$: temperature change

- α is **material-dependent** and **tabulated** (see next slide).
- $1\,\frac{1}{\mathrm{K}} = 1\,\frac{1}{{}^\circ\mathrm{C}}$ for *changes* in temperature (the size of a kelvin = the size of a Celsius degree).

Coefficients of Linear Expansion

Material	$\boldsymbol{\alpha}$ $\left(\times 10^{-6}\ \frac{1}{^\circ\mathrm{C}}\right)$
Aluminum	25
Brass	19
Copper	17
Steel / Iron	12
Concrete	12
Glass (ordinary)	9
Pyrex glass	3
Invar (Ni-Fe alloy)	0.9

Example: Aluminum Rod

A piece of aluminum rod has length $L_1 = 2.000\,\mathrm{m}$ at $T_1 = 17\,^\circ\mathrm{C}$.
It is heated to $T_2 = 97\,^\circ\mathrm{C}$. Find its final length.
($\alpha_{\mathrm{Al}} = 25 \times 10^{-6}\,\frac{1}{^\circ\mathrm{C}}$)

$$\Delta L = \alpha L_1 \Delta T$$

$$\Delta L = 25 \times 10^{-6}\,\frac{1}{^\circ\mathrm{C}}(2.000\,\mathrm{m})(80\,^\circ\mathrm{C})$$

$$\Delta L = 0.004\,00\,\mathrm{m} = 4.00\,\mathrm{mm}$$

$$L_2 = L_1 + \Delta L$$

$$\boxed{L_2 = 2.004\,\mathrm{m}}$$

$\Delta T = 97 - 17 = 80\,^\circ\mathrm{C}$

A 4 mm change in a 2 m bar — this is small but significant in precision engineering.

Example: Golden Gate Bridge

The main span of the Golden Gate Bridge is $L = 1275\,\text{m}$.
San Francisco's temperature ranges from $-15\,°\text{C}$ to $40\,°\text{C}$.
How much does the bridge expand between winter and summer?
$(\alpha_{\text{steel}} = 12 \times 10^{-6}\,\frac{1}{°\text{C}})$

$$\Delta T = 40 - (-15) = 55\,°\text{C}$$

$$\Delta L = \alpha L \Delta T$$

$$\Delta L = 12 \times 10^{-6}\,\frac{1}{°\text{C}}\,(1275\,\text{m})(55\,°\text{C})$$

$$\boxed{\Delta L \approx 0.84\,\text{m}}$$

The bridge changes length by nearly **1 meter**. This is why expansion joints are built into all large structures.

Check Your Understanding

A steel railroad track is 30.0 m long at 15.0 °C.

(a) How much does it expand if the temperature rises to 50.0 °C?

(b) If tracks are laid end-to-end with no gaps, what problem arises in summer?

$(\alpha_{\text{steel}} = 12 \times 10^{-6} \, \frac{1}{°\text{C}})$

Come to lecture to see the answer.

Check Your Understanding

A circular hole is drilled through a metal plate.

When the plate is heated, does the hole get **larger** or **smaller**?

Explain your reasoning.

Come to lecture to see the answer.

Outline of Part II: Thermal Expansion

Linear Expansion

Volume Expansion

Area and Volume Expansion

Area Expansion — $\Delta A = 2\alpha A \Delta T$

A surface that expands linearly in each dimension by $\alpha \Delta T$ expands in area by approximately $2\alpha \Delta T$:

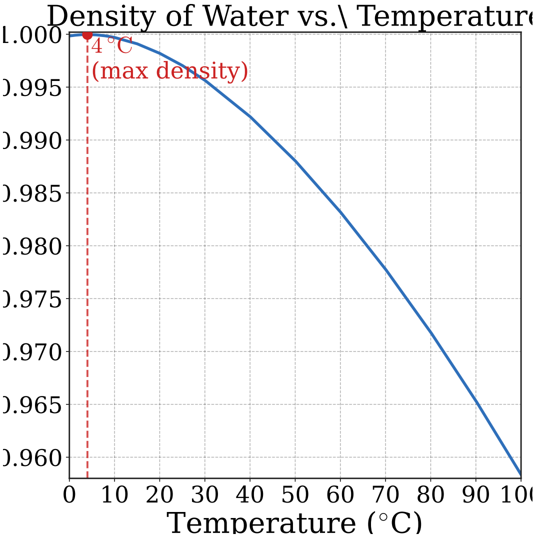
$$\Delta A = 2\alpha A \Delta T$$

Volume Expansion — $\Delta V = \beta V \Delta T$

$$\Delta V = \beta V \Delta T \qquad \text{where} \quad \beta \approx 3\alpha$$

- β: coefficient of volume expansion $\left(\frac{1}{^\circ\mathrm{C}}\right)$
- Liquids and gases have much larger β than solids.

Water's Anomalous Expansion



Why This Matters

- Lakes freeze from the *top* down, not the bottom up.
- Ice layer insulates water below, allowing aquatic life to survive winter.
- If water behaved normally, lakes would freeze solid and kill all life.
- Water expanding on freezing can crack rocks, burst pipes, and damage cells in frozen biological tissue.

Water is Unusual

Example: Gasoline Tank Overflow

A steel tank holds 60.0 L of gasoline at 15 °C.
How much gasoline overflows if the temperature rises to 35 °C?
$(\beta_{\text{gasoline}} = 950 \times 10^{-6}\,\frac{1}{°\text{C}},\ \beta_{\text{steel}} = 36 \times 10^{-6}\,\frac{1}{°\text{C}})$

$$\Delta V_{\text{gas}} = 950 \times 10^{-6}\,\frac{1}{°\text{C}}\,(60.0\,\text{L})\,(20\,°\text{C}) = 1.14\,\text{L}$$

$$\Delta V_{\text{tank}} = 36 \times 10^{-6}\,\frac{1}{°\text{C}}\,(60.0\,\text{L})\,(20\,°\text{C}) = 0.0432\,\text{L}$$

$$\boxed{V_{\text{overflow}} = 1.14\,\text{L} - 0.0432\,\text{L} \approx 1.10\,\text{L}}$$

Gasoline expands much more than steel ($\beta_{\text{gas}} \gg \beta_{\text{steel}}$), which is why fueling on a hot day is inefficient — you pump the same mass but in a larger volume.

Part III

The Ideal Gas Law

Outline of Part III: The Ideal Gas Law

Outline of Part III: The Ideal Gas Law

Development of the Ideal Gas Law

Applications of the Ideal Gas Law

Effect of Temperature on Gases at Constant Volume

Isochoric Process ($\Delta V = 0,\ V =$ const.)

If temperature increases (more molecular KE):

- $\bar{v}$ increases $\Rightarrow$ molecules hit walls harder and more often
- $\Rightarrow$ force F on wall increases $\Rightarrow$ pressure P increases

$$\boxed{P \propto T} \qquad (V \text{ and } N \text{ held constant})$$

This is why car tires have higher pressure after driving (friction heats the air inside at nearly constant volume).

Low T **High** T

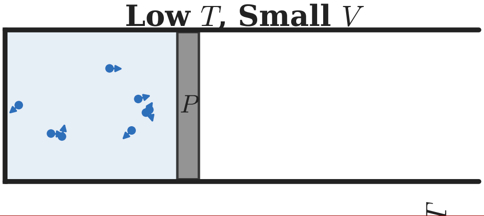
Effect of Temperature on Gases at Constant Pressure

Isobaric Process ($\Delta P = 0$, $P =$ const.)

If V is free to change and T increases:

- Molecules push outward harder $\Rightarrow$ volume expands until equilibrium is restored
- $\Delta V = A \Delta h$ (piston moves out)

$$\boxed{V \propto T} \qquad (P \text{ and } N \text{ held constant})$$



Work is done by the gas as it expands against the atmosphere:
$W = P\Delta V = F\Delta h.$

Effect of Adding More Gas at Constant Temperature and Pressure

Constant T and P: Varying N

Suppose $\Delta P = 0$ and $\Delta T = 0$ (a balloon of air).

As you blow more air in (N increases), the balloon expands to maintain the same pressure:

$$\boxed{V \propto N} \qquad (P \text{ and } T \text{ held constant})$$

Combining All Three Proportionalities

$$P \propto T, \quad V \propto T, \quad V \propto N \qquad \Rightarrow \qquad PV \propto NT$$

Experiments determined the proportionality constant to be $k_B = 1.38 \times 10^{-23}\,\frac{\text{J}}{\text{K}}$ (Boltzmann constant):

The Ideal Gas Law

Molecular Form

$$PV = Nk_BT$$

- P: absolute pressure $(\mathrm{Pa} = \mathrm{N/m^2} = \mathrm{J/m^3})$
- V: volume $(\mathrm{m^3})$
- N: number of molecules (dimensionless)
- $k_B = 1.38 \times 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}}$: Boltzmann constant
- T: temperature in kelvin

Molar Form

Moles, Molecules, and Avogadro's Number

Avogadro's Number

$$N_A = 6.022 \times 10^{23}\ \frac{1}{\text{mol}}$$

A **mole** is N_A of something — just as a **dozen** is 12 of something.

Converting n to N

$$N = n \cdot N_A$$

$$n = N/N_A$$

The two gas law forms are equivalent:

$$R = k_B \cdot N_A$$

How Big is a Mole?

- One mole of pennies stacked would reach the Moon and back about 6×10^{14} times.
- One mole of water molecules = 18.0 g of water.
- All gases at STP: 1 mole occupies

Outline of Part III: The Ideal Gas Law

Development of the Ideal Gas Law

Applications of the Ideal Gas Law

The Combined Gas Law

Relating Two States of the Same Gas

For a fixed amount of gas ($n_1 = n_2$, no leaks), applying $PV = nRT$ to two states:

$$\frac{P_1 V_1}{n_1 T_1} = R = \frac{P_2 V_2}{n_2 T_2} \qquad \Longrightarrow \qquad \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

- Special cases: hold one variable constant to recover Boyle's, Charles's, or Gay-Lussac's law.
- Always convert temperature to Kelvin before substituting.
- Pressure may be in any consistent unit (Pa, atm, mmHg) as long as P_1 and P_2 use the same unit.

Example: Bicycle Tire Pressure

A bicycle tire has gauge pressure 7.00×10^5 Pa when inflated at $18.0\,°\mathrm{C}$ in a garage. The rider parks outside where $T = 35.0\,°\mathrm{C}$. What is the new gauge pressure? (Assume the tire volume does not change; $P_{\mathrm{atm}} = 1.013 \times 10^5$ Pa)

$$P_1 = P_g + P_{\mathrm{atm}} = (7.00 + 1.013) \times 10^5 = 8.013 \times 10^5\,\mathrm{Pa}$$

$$T_1 = 18.0 + 273.15 = 291.15\,\mathrm{K}$$

$$T_2 = 35.0 + 273.15 = 308.15\,\mathrm{K}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad (V\ \mathrm{const.})$$

$$P_2 = P_1 \cdot \frac{T_2}{T_1} = 8.013 \times 10^5\,\mathrm{Pa} \cdot \frac{308.15}{291.15}$$

$$P_2 = 8.48 \times 10^5\,\mathrm{Pa}$$

New gauge pressure:

$$\begin{aligned} P_{g,2} &= P_2 - P_{\mathrm{atm}} \\ &= 8.48 - 1.013 \end{aligned}$$

$$\boxed{P_{g,2} \approx 7.47 \times 10^5\,\mathrm{Pa}}$$

A temperature increase of only $17\,°\mathrm{C}$ raises tire pressure by abo 0.47×10^5 Pa.

Example: Balloon in the Freezer

A balloon has $V_1 = 1.00\,\mathrm{L}$ at $T_1 = 25\,°\mathrm{C}$ and $P_1 = 1.00\,\mathrm{atm}$.
It is placed in a freezer at $T_2 = 0\,°\mathrm{C}$. What is its new volume? ($P_2 = P_1$, sealed balloon.)

$$T_1 = 25.0 + 273.15 = 298.15\,\mathrm{K}$$

$$T_2 = 0.00 + 273.15 = 273.15\,\mathrm{K}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = V_1 \cdot \frac{T_2}{T_1} \cdot \frac{P_1}{P_2}$$

$$V_2 = 1.00\,\mathrm{L} \cdot \frac{273.15}{298.15} \cdot 1$$

The balloon shrinks by about 8% as it cools.

Temperature *must* be in kelvin:

- if we used Celsius: $0/25 = 0$ (wrong!)
- in Kelvin: $273.15/298.15 = 0.916$ (correct)

Check Your Understanding

A gas occupies $2.00\,\mathrm{L}$ at $300\,\mathrm{K}$ and $1.00\,\mathrm{atm}$.

(a) What is its volume at $600\,\mathrm{K}$ and $1.00\,\mathrm{atm}$?

(b) What is its volume at $300\,\mathrm{K}$ and $2.00\,\mathrm{atm}$?

Come to lecture to see the answer.

Check Your Understanding

Estimate the number of moles of air in a normal breath.

Assume: $V_{\text{breath}} \approx 2.00\,\text{L}$, body temperature $T = 37.0\,°\text{C}$, $P = 101\,325\,\text{Pa}$.

Come to lecture to see the answer.

Part IV

Kinetic Theory

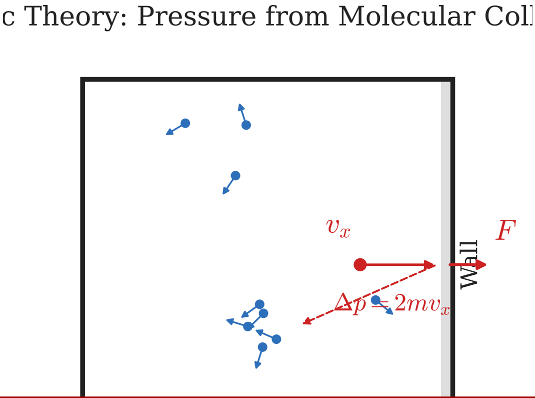
Outline of Part IV: Kinetic Theory

Outline of Part IV: Kinetic Theory

Kinetic Theory: Pressure from Molecular Collisions

Gas pressure arises from molecules **colliding with the walls** of their container.

Each collision imparts a small impulse Δp to the wall; summed over $\sim 10^{23}$ molecules, this produces a macroscopic force.



- Higher $T \Rightarrow$ higher $\overline{v^2} \Rightarrow$ harder collisions $\Rightarrow$ higher P.
- More molecules $(N) \Rightarrow$ more collisions $\Rightarrow$ higher P.

Temperature and Kinetic Energy

Connecting $PV = Nk_BT$ with $PV = \frac{1}{3}Nm\overline{v^2}$

Setting the two expressions for PV equal:

$$Nk_BT = \frac{1}{3}Nm\overline{v^2} \qquad \Rightarrow \qquad \frac{3}{2}k_BT = \frac{1}{2}m\overline{v^2}$$

Average Kinetic Energy per Molecule

$$\overline{KE} = \frac{1}{2}m\overline{v^2} = \frac{3}{2}k_BT$$

- Temperature is **directly proportional to** average kinetic energy per molecule.
- All gases at the same temperature have the *same* average kinetic energy, regardless of molecular mass.

Root-Mean-Square Speed

v_{rms} — the Typical Molecular Speed

The **root-mean-square speed** is the square root of the average of v^2:

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_B T}{m}}$$

where m is the mass of one molecule.

- Heavier molecules move **slower** at the same T (since $\overline{KE}$ is the same, but m is larger, v must be smaller).
- Lighter molecules move **faster** at the same T.
- For molar quantities: $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$ where M is molar mass in $\frac{\text{kg}}{\text{mol}}$.

Example: Kinetic Energy and Speed of N_2 at Room Temperature

Find (a) the average kinetic energy and (b) the rms speed of a nitrogen molecule (N_2) at 20.0 °C.
($m_{N_2} = 28.0\,\frac{\text{g}}{\text{mol}}$)

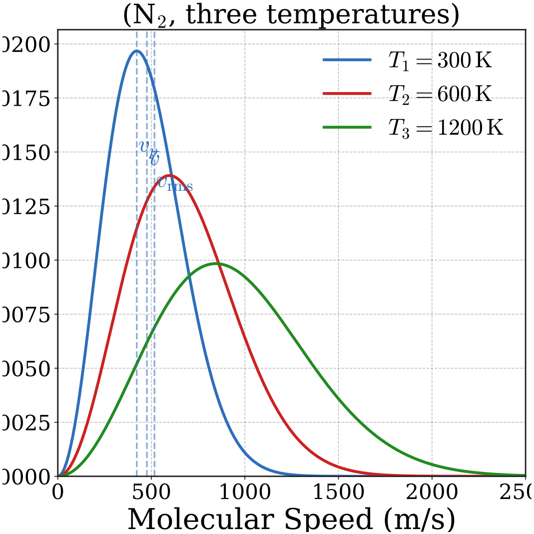
$$T = 20.0 + 273.15 = 293.15\,\text{K}$$

(a)
$$\overline{KE} = \frac{3}{2} k_B T$$
$$= \frac{3}{2}\left(1.38 \times 10^{-23}\,\frac{\text{J}}{\text{K}}\right)(293.15\,\text{K})$$
$$\boxed{= 6.07 \times 10^{-21}\,\text{J}}$$

(b)
$$M = 0.0280\,\frac{\text{kg}}{\text{mol}}$$
$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$
$$= \sqrt{\frac{3(8.314)(293.15)}{0.0280}}$$
$$\boxed{\approx 511\,\frac{\text{m}}{\text{s}}}$$

Outline of Part IV: Kinetic Theory

The Maxwell-Boltzmann Distribution



Effect of Temperature

- Higher T: curve shifts to the *right* and flattens (wider distribution).
- Lower T: curve shifts left and narrows.
- Heavier molecules: narrower curve at same T.
- Lighter molecules: broader, faster distribution.

The high-speed *tail* of the distribution explains evaporation (fast molecules escape) and chemical reactions (only

Check Your Understanding

At the same temperature, which gas molecules have **higher average kinetic energy**: N_2 (molar mass $28\,\frac{\text{g}}{\text{mol}}$) or O_2 (molar mass $32\,\frac{\text{g}}{\text{mol}}$)?

Which molecules have **higher** v_{rms}?

Come to lecture to see the answer.

Check Your Understanding

Calculate the rms speed of an oxygen molecule (O_2) at body temperature $T = 37.0\,°\mathrm{C}$.

$(M_{O_2} = 32.0\,\frac{\mathrm{g}}{\mathrm{mol}})$

Come to lecture to see the answer.

Part V

Phase Changes and Humidity

Outline of Part V: Phase Changes and Humidity

Phase Changes

Humidity, Evaporation, and Boiling

Outline of Part V: Phase Changes and Humidity

Phase Changes

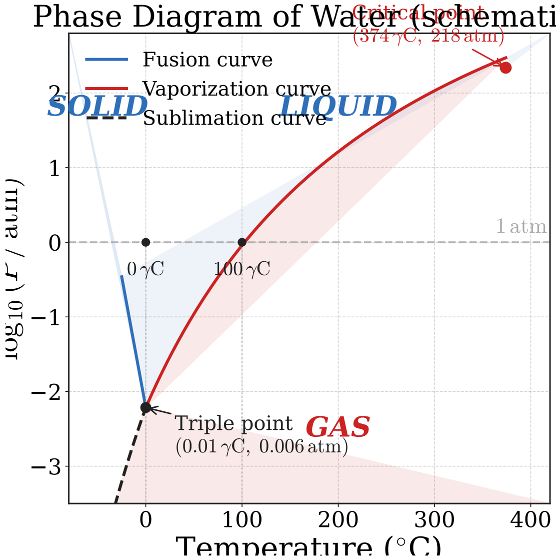
Humidity, Evaporation, and Boiling

States of Matter

Property	**Solid**	**Liquid**	**Gas**
Shape	Fixed	Takes container shape	Takes container shape
Volume	Fixed	Fixed	Fills container
Compressibility	Negligible	Negligible	High
Molecular order	High (lattice)	Moderate	None
Molecular spacing	Very close	Close	Far apart

Phase transitions occur when enough energy is added (or removed) to change molecular arrangement. Temperature *does not change* during a phase transition —

The Phase Diagram



Key Features

- **Triple point**: all three phases coexist in equilibrium. For water: $T = 0.01\,°\mathrm{C}$, $P = 0.006\,00\,\mathrm{atm}$.
- **Critical point**: above this T and P, no distinct liquid-gas boundary.
- **Sublimation curve**: solid $\leftrightarrow$ gas (no liquid phase).
- Normal boiling/melting points: where curves cross $P = 1\,\mathrm{atm}$.

Sublimation and Pressure Dependence of Boiling Point

Sublimation

- At pressures below the triple point, a substance transitions directly from solid to gas.
- **Dry ice** (solid CO_2): sublimes at $-78.5\,°C$ at 1 atm. CO_2 cannot be liquid at atmospheric pressure.
- **Freeze drying**: food frozen below triple point; pressure reduced; water sublimes, leaving dry food.

Pressure Cooker

- Boiling point *increases* with pressure.
- A pressure cooker at 2 atm raises water's boiling point to $\approx 120\,°C$.
- Hotter water cooks food faster.
- Conversely, water boils at $90\,°C$ at high altitude (low atmospheric pressure), so cooking takes longer.

Check Your Understanding

(a) Why does condensation (e.g. dew) form most readily on the **coldest** surfaces in a room?

(b) Can carbon dioxide (CO_2) be liquefied at room temperature ($20\,^\circ C$) by applying pressure?
Critical temperature of CO_2: $31.1\,^\circ C$.

Come to lecture to see the answer.

Outline of Part V: Phase Changes and Humidity

Evaporation and Vapor Pressure

Evaporation

Fast molecules at a liquid surface escape into the gas phase even *below* the boiling point.

The boiling point is the temperature at which vapor pressure equals atmospheric pressure, allowing bubble formation throughout the liquid — not just at the surface.

Vapor Pressure

- The pressure exerted by vapor in equilibrium with its liquid.
- Increases rapidly with temperature.
- Independent of the presence of air.

Why Water Boils in a Vacuum

Even at room temperature, water in a sealed chamber will boil if the pressure is reduced below its current vapor pressure. The liquid has enough molecular speed to escape at low enough external pressure.

Relative Humidity

Relative Humidity

$$\%_{\text{RH}} = \frac{\rho_{\text{vap}}}{\rho_{\text{sat.vap}}} \times 100\%$$

- ρ_{vap}: actual water vapor density in the air (g/m^3)
- $\rho_{\text{sat.vap}}$: saturation vapor density at that temperature (g/m^3, tabulated)
- At 100 % RH: air is **saturated** — evaporation and condensation rates balance; no net evaporation.

- The **dew point** is the temperature at which the current humidity would be 100% (air cools to saturation, dew forms).
- Hot, humid days feel worse than hot, dry days because high RH inhibits

Saturation Vapor Density vs. Temperature

Temperature (°C)	**Saturation Vapor Density** ($\frac{\text{g}}{\text{m}^3}$)
-20	0.89
0	4.85
10	9.40
20	17.2
30	30.4
37	44.0

Saturation vapor density increases dramatically with temperature. A warm room can "hold" far more water vapor than a cold room. This is why heated air is dry in winter: outside air at 0 °C contains little vapor; when warmed to 20 °C indoors, its

Example: Calculating Relative Humidity

On a 20 °C day, the air contains water vapor with density $\rho_{\text{vap}} = 9.40\,\frac{\text{g}}{\text{m}^3}$. What is the relative humidity? (Saturation vapor density at 20 °C: $17.2\,\frac{\text{g}}{\text{m}^3}$.)

$$\%_{\text{RH}} = \frac{\rho_{\text{vap}}}{\rho_{\text{sat.vap}}} \times 100\% = \frac{9.40\,\frac{\text{g}}{\text{m}^3}}{17.2\,\frac{\text{g}}{\text{m}^3}} \times 100\%$$

$$\boxed{\%_{\text{RH}} = 54.7\%}$$

This is typical comfortable indoor humidity. Above $\sim$70%, air feels muggy; below $\sim$30%, the air feels dry and can cause skin irritation and static electricity.

Check Your Understanding

(a) Why does sweating cool the body?

(b) Why is a $35\,^{\circ}\mathrm{C}$ day with $95\,\%$ RH more dangerous than a $38\,^{\circ}\mathrm{C}$ day with $20\,\%$ RH?

Come to lecture to see the answer.

Check Your Understanding

Why does rubbing alcohol evaporate much more quickly than water at the same temperature?

(Hint: consider the intermolecular forces involved.)

Come to lecture to see the answer.

Chapter 13 Summary

Quantity / Law	**Equation**
Temperature conversion (F ↔ C)	$T_F = 1.8\,T_C + 32$
Temperature conversion (C ↔ K)	$T_K = T_C + 273.15$
Linear thermal expansion	$\Delta L = \alpha L \Delta T$
Area thermal expansion	$\Delta A = 2\alpha A \Delta T$
Volume thermal expansion	$\Delta V = \beta V \Delta T, \quad \beta \approx 3\alpha$
Ideal gas law (molecular)	$PV = Nk_B T$
Ideal gas law (molar)	$PV = nRT$
Combined gas law	$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$
Molecular pressure	$PV = \frac{1}{3} N m \overline{v^2}$